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AI and Research Methods

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10.1 INTRODUCTION

In this chapter, we survey current uses of artificial intelligence (AI) in political science and consider how these tools are reshaping political methodology. A brief note on terminology is useful at the outset. Throughout, we use "AI" primarily to refer to large language models (LLMs), including models developed by providers such as OpenAI, Anthropic, Meta, and DeepSeek.

To orient the reader, the chapter's main contributions fall into six substantive domains, each of which highlights a distinct way that LLMs are beginning to reshape political science research. Section 10.2 examines annotation and measurement, showing how LLMs can be used to classify and extract information from unstructured political data, while also arguing that their use requires new approaches to validation when human coding is no longer an unquestioned gold standard.

Section 10.3 turns to treatment generation, not only demonstrating how LLMs can expand the design space of experiments by making it easier to generate and tailor stimuli but also emphasizing the need to guard against bundled treatments and to validate construct fidelity.

Section 10.4 addresses silicon sampling, or the use of LLMs to generate synthetic proxies for human responses, and shows that the promise of this approach depends fundamentally on matching validation strategies to the inferential purpose of the synthetic data.

Section 10.5 considers generative agent-based modeling, explaining how LLMs can replace hand-coded behavioral rules with richer, language-mediated

Some authors of this chapter used AI tools, including OpenA GPT 5.3, Codex 5.2, and Claude Code Opus 4.6 to: identify typographic and flow errors, search for citations, reformat structure, redraft sections, and draft initial sections. Each author or author pair was responsible for their respective sections. Christopher Barrie wrote the Introduction and Discussion as well as sections 10.5 and 10.8; Jon Mellon and Margaret Roberts wrote section 10.2; Christopher Lucas and Alexis Palmer wrote section 10.3; Lisa Argyle wrote section 10.4; Michael Heseltine wrote section 10.6; and James Bisbee and Arthur Spirling wrote section 10.7.

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agent policies, while also introducing new challenges of specification, interpretability, and empirical validation.

Section 10.6 examines tool-augmented data collection, focusing on the use of LLMs as research agents that can search, retrieve, parse, and organize information across heterogeneous sources. Its central contribution is to show how these systems can lower the cost of data assembly while making provenance, logging, and source verification newly important.

Section 10.7 then shifts perspective and treats LLMs not as instruments but as objects of substantive inquiry. It outlines how political scientists can audit and evaluate models themselves, whether to understand their usefulness as research tools, their social and political effects, or their broader implications for how we think about agency, reasoning, and bias.

Each section is written by one or two contributors selected for their expertise in the relevant domain. Unsurprisingly, the authors do not agree on every aspect of how AI should be used in political science. That lack of full consensus reflects the present moment: one of rapid change, methodological experimentation, and genuine uncertainty about the consequences of these emerging technologies. This lack of consensus is perhaps most obvious when it comes to questions of replication standards when using LLMs and the implications of LLMs for transparency and reproducibility more generally. Drawing on Sections 10.2–10.7, we identify both the emerging points of agreement and the open methodological questions that remain.

10.2 LLMS FOR ANNOTATION AND MEASUREMENT

10.2.1 The Task

Political science studies people, bureaucracies, and governments that produce large quantities of texts, images, and videos. From political speeches to bureaucratic agreements to social media videos, political scientists often need to distill and process information from unstructured data. While historically political scientists would distill these texts, images, and videos themselves, with the help of research assistants or crowd-sourcing, increasingly this processing can be done more efficiently and cheaply by AI.

In this chapter, we explore both the promise and pitfalls of using LLMs to label or measure something from unstructured data for further downstream analysis. The task we take on in this chapter is simple: A political scientist wants to take a dataset of images, texts, or videos and assign them to categorical or numerical values. They then want to use the outputs in downstream analysis, for example, to describe the whole dataset or to connect the extracted measure with other metadata.

We emphasize that using LLMs for this task still requires human validation and, in what follows, we give guidance on the choices researchers must make while using LLMs for annotating and for deciding how best to validate their

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accuracy. In addition, we surface considerations – from personalization to replication to training data – that are unique to LLMs and merit further discussion in the field.

10.2.2 Where We Came From

Labeling with LLMs has two distinct lineages: human research assistants and traditional machine learning and statistical methods. Since the start of political science, researchers have wanted to convert unstructured text such as manifestos or political speeches into quantifiable categories. For instance, Lasswell (1941) coded newspaper headlines by whether they mentioned the US. For most tasks this has simply involved having human beings (either the researcher themselves or their students) classify the text into categories. Here, researchers are expected to have a clear definition of their measure, good coding rules, and high intercoder reliability among human labelers (Neuendorf 2016).

While substantial work has gone into trying to make such labels reliable, the baseline reproducibility across humans is not typically high, with Kappa scores of 0.8 being referred to as "almost perfect" despite meaning that 10 percent of cases would be coded differently (for a binary metric with prevalence of 50 percent). Against this backdrop, LLM use cases that replace human workflows often seem like an improvement in reproducibility.

More recently, human labelers have been used as training datasets for machine-learning algorithms and classifiers to more systematically label larger sets of data that could not be practically coded by humans. In these settings, the researcher trains their own machine-learning model to replicate the human-classified labels with sufficiently accurate performance. For these approaches, researchers seek high out-of-sample precision and recall in retrieving human-classified labels. These approaches also have high levels of reproducibility because code can be provided. However, they still rely on human-classified labels, which are less reproducible and can have errors. Machine-coded classifiers can also have systematic biases and may not be generalizable to other datasets.

10.2.3 What This Method Does

Large language models generate text by predicting the next sequence of text from a text input. They can, therefore, be used to generate responses to questions such as, "On a scale of 1–10, 10 being most positive, how positive is the following text? 'It's a beautiful day!' Please provide only a number as output." In this way, researchers are using LLMs to label texts by providing specific instructions and codebooks to LLMs and evaluating many texts in sequence.

Large language models are "off-the-shelf" in that, unlike previous classifiers trained by the researcher, they are pretrained by large corporations or research groups. There are now a vast array of potential LLMs for social scientists to

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consider using and many different options within each one. These choices will have impacts on the performance and cost of many annotation tasks. Our broad advice is that researchers empirically measure both the cost and the performance of LLMs on a task and pick the cheapest option that achieves what they consider to be an acceptable level of performance. We strongly recommend that researchers think carefully about what performance gains are worth significant cost increases. While more expensive models and settings often perform better than cheaper ones, this is not universal and should be tested for a scientist's exact use case. We talk more about validation steps later in this chapter.

To give an idea of just how much cost difference there can be, consider a scenario where the task is to label sentiment on social media platforms (aka "tweets") using a long codebook that takes up 5,000 tokens and the typical tweet is just 50 tokens.

- Option A: GPT-5.2 with high reasoning and one tweet passed at a time and no attempt to use batch processing or prompt caching.
- Option B: GPT-5-nano with no reasoning and 100 tweets passed at a time using the batch processing Application Programming Interface (API).

For Option A, a researcher would likely spend around $149,000 to code one million tweets. For Option B, they would spend $2.90–50,000 times cheaper (or, to put this another way, one could consider coding 50 billion tweets instead of 1 million for the same cost). It may be that Option B does not produce acceptable performance levels for a research question, but, with hundreds of thousands of dollars on the line, it is probably worth checking.[1]

The first choice is simply which provider and/or API to use. At time of writing, Open AI, Google, and Anthropic are close in price and performance on most benchmarks. There are also open-weight models to consider. These have the advantage of allowing full reproducibility because the exact model artifact can be stored, whereas closed-source providers may deprecate models or withdraw access to older versions over time (Barrie, Palmer, and Spirling 2025).

Open-weight models can be run on dedicated cloud providers such as replicate.com and openrouter.com. This makes their use largely similar to the closed weight providers except that it is usually possible to record the specifics of the analysis in enough detail that a result should be fully reproducible. However, the downside of this is that cloud providers of open-weight models typically offer little cost savings over using a closed weight model (while usually falling short on performance at a given price point). Another option for open-weight models is for researchers to run them on hardware they control. For smaller models this can be possible even on a standard laptop (Apple's recent

[1] The cost differences here come from a 25× difference in token cost, a 50 percent flex discount, saving nearly 99 percent of input token costs through input batching, and a 10× savings from avoiding the use of reasoning tokens.

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Macbook computers are well-suited to this due to the high available memory for GPU processing). The setup for this is still relatively straightforward and can be managed through software such as Ollama and LM studio. For larger models, users typically need access to substantial computing infrastructure. These types of computing resources are not available across all universities, limiting this option to significantly fewer departments in practice.

The LLM landscape is changing fast, so the various cost/performance/reproducibility trade-offs will likely continue to change.

After deciding on a model to use, how should a researcher use it? While many people will be most familiar with LLMs through chat interfaces such as ChatGPT, we strongly recommend against using these interfaces for research purposes. Chat interfaces are opaque in what model is being called with what settings, are not well suited for large-scale tasks, and create room for copy-paste errors. Chat interfaces also create personalization, which means that the output can be specific to the history of a person's queries, which is not good for generalizability. We suggest that researchers use the APIs of the models for efficiency.

After choosing a model, the most important task is developing an effective prompt to ask the LLM to complete the labeling. In our experience, many initial LLM (and RA) failures result from having poorly described the task rather than the model being incapable of completing it. While early guidance suggested various tactics from telling the model to adopt a persona to threatening the model, the consensus has generally shifted toward clarity of the task, giving examples and iteratively improving the prompt. Large language models can be particularly susceptible to prompts that lead the LLM to produce a desired result, therefore, researchers should pay particular attention to the nature of the prompt (Anagnostidis and Bulian 2024).

Outside of the prompt, most LLM APIs will give several parameters that can be set that governs the output of the model. While a full description of all settings would be long and quickly out of date, two parameters are important to pay attention to: temperature and reasoning.

At the final layer of an LLM neural network, the model produces a score for each possible next token. These scores are translated into probabilities of choosing a token using a softmax function. This function can be tuned with a temperature parameter to spread the probabilities more evenly across potential next tokens (high temperature) or concentrate the probabilities on fewer tokens (low temperature). At the extreme, setting temperature to zero will typically make the output of an LLM deterministic by always picking the highest probability next token (though there are some nuances that mean this is not quite true in practice). In general, users should set the temperature to zero if the goal is to get close to reproducible labels. There are some circumstances where a scientist may want to choose a higher temperature if the variation across possible LLM responses is of interest (for instance, using multiple generations to find a consensus label).

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Reasoning is a newer parameter. Traditionally, LLMs have simply generated the next tokens based on the input text fed through the neural network. However, AI researchers soon realized that LLMs could achieve better results for certain problems when told to "think step by step" (Wei et al. 2022). This has now been integrated into LLM architecture through an internal reasoning trace where LLMs first generate tokens that represent "reasoning" about a problem and then generate a final answer based on the input text and the reasoning steps that they wrote out.

An important factor for researchers to consider is that reasoning tokens are charged at the output token rate, which is typically around ten times the cost of input tokens. This means that the cost of reasoning can easily exceed the cost of the input text. Most models that support reasoning allow some level of control of the reasoning length either through a reasoning effort parameter (e.g., Open AI) or a reasoning token budget (e.g., Google). As usual, we recommend choosing the lowest level that achieves the desired performance.

One further choice that researchers should make is how many cases they pass to the LLM each time. Most people simply pass a set of instructions followed by a single case to label. However, this is cost inefficient because providers charge for passing the whole fixed instruction set (which can often be thousands of words) every time.

Some providers offer substantial discounts for repeated prompt prefixes, and asynchronous batch processing can also materially reduce costs. Another option is to pass more than one case at a time (input batching) (Cheng et al. 2023). For instance, Mellon et al. (2024) pass twenty-five open text responses at a time to the model. This spreads the cost of a long set of instructions over a large set of inputs. Importantly, using input batching can be combined with flex or batch processing discounts from providers that typically offer a 50 percent discount in exchange for a delay in processing the request. For many real-world tasks, input batching can achieve cost savings of greater than 90 percent. As always, it is important to validate performance when using these methods. Regardless of what choices a researcher makes, it is vital to document all decisions. Importantly, this includes choices that have default values. Researchers must record all parameters because it is not guaranteed that it will be possible to reconstruct what these were after the fact.

While much research in this area has used LLMs "off-the-shelf," many models have the option of inserting data that the researcher has labeled themselves to "fine-tune" the model. While fine-tuning requires significant human effort to create the training data, it can have substantial performance improvements for some tasks (Wang 2024; Waight et al. 2025).

Another important consideration when labeling data is data handling. Norms and interpretations of laws are not solidified in this area but, at a minimum, researchers should be careful that they do not put personally identifiable information into systems that will use that data for training. There are verified examples of LLMs reproducing training data in outputs (Kiyomaru et al.

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2024), so a conservative assumption is that any data used to train an LLM has a small probability of being reproduced in its entirety in future. Large language model terms and conditions are complex and subject to change, so we recommend getting advice from within your institution.

While the most common uses of LLMs is for labeling text data, recent models can ingest other forms of information such as images, audio, or video. For example, this is an area that may be particularly fruitful for campaign studies, where LLMs could, for instance, classify advertisements with visual elements at scale. The best practices for labeling this kind of information are largely similar to labeling text, although we do recommend being mindful of the costs of using this kind of media, which can be high in comparison to simple text inputs.

Given the wide range of choices, researchers face when using LLMs, we strongly recommend testing a task using multiple models, settings, and parameters to find an option that represents a good balance between cost and performance.

10.2.4 Example Applications

Recent work illustrates both the promise and the range of LLM-based annotation in political science. Ornstein and colleagues (2025) provide one of the clearest general demonstrations. They revisit several established political text-analysis tasks and show that few-shot prompting can often substitute for more labor-intensive human labeling or conventional supervised pipelines, while requiring much less labeled training data and dramatically lower cost. One especially strong validation example comes from Wu, Nagler, Tucker, and Messing (2023), who use ChatGPT-3.5 to estimate the latent positions of US senators using pairwise comparisons and Bradley–Terry scaling. In that setting, they report very strong agreement with existing ideology benchmarks, suggesting that LLMs can recover meaningful latent structure when the objects being coded are highly salient and richly represented in public text.

At the same time, the strongest results so far often come from relatively favorable validation settings, and that should make researchers cautious about over-generalizing. The Wu et al. (2023a). application, for example, focuses on contemporary US senators – a high-profile set of political actors for whom both background information and external validation measures are unusually abundant. Likewise, Heseltine and Clemm von Hohenberg (2024) show that GPT-4 can perform very well when coding political text across multiple countries and tasks, comparing GPT-4 against human expert coding of tweets and news articles in the United States, Chile, Germany, and Italy. They report especially high accuracy on shorter texts such as tweets, with weaker performance on longer news articles and slightly lower performance outside English; they also show that a hybrid workflow, in which humans adjudicate disagreements across multiple GPT-4 runs, can improve reliability further.

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These are impressive findings, but they also suggest a boundary condition: current evidence is strongest for relatively visible domains, shorter texts, and tasks where good benchmarks already exist. That means researchers should be careful in extending these successes to lower-salience politicians, less-covered institutions, lower-resource languages, historical corpora, or more weakly structured classification problems, where training-data coverage and independent validation may be much thinner.

Information retrieval tasks represent an adjacent field in the use of LLMs for annotation or data generation tasks. Mellon et al. (2024) show that language models can be used to code open-ended survey responses about the "most important issue" into substantive categories at scale, extending LLM-based annotation beyond elite political text and into survey research. Bol and Bono (2025), meanwhile, show that ChatGPT can estimate party positions in France with reasonable success when compared against contemporaneous expert judgments, particularly on major and salient dimensions. Taken together, these studies make a strong case that LLM-based annotation is already a useful methodological tool, especially when researchers need scalable coding at low cost. But they also imply that the next step is not simply to accumulate more success stories in easy cases. The more important challenge is to determine how well these methods generalize to harder settings – where labels are noisier, texts are longer or less standardized, actors are less prominent, and gold-standard validation data are limited.

10.2.5 Validation and Inference

Validation is critical when using LLMs for annotation because performance varies between datasets and tasks. Large language model outputs should be compared to human-coded data as validation that it is, indeed, measuring the concept as interpreted by humans. Like other machine learning workflows, performance should be evaluated "out of sample," that is, not using examples that have already been provided to the LLM or used in fine-tuning.

Many studies simply use agreement rates between LLMs and humans as the main metric to assess coder performance. However, this can be problematic for categories of rare events (e.g., hate speech in tweets) as a model can perform well simply by always choosing the largest category.

We recommend the use of balanced outcome metrics such as F1 (the harmonic mean of recall and precision) to reflect both types of performance. Other options include directly examining the confusion matrix.

One additional complication is that types of error matter more for different applications. False positives are especially problematic when the goal of the method is to characterize the population of positives in a low prevalence context. For instance, suppose that an LLM falsely classifies a tweet as hate speech 2 percent of the time but that the prevalence of hate speech tweets is only half of 1 percent.

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Researchers should also consider that LLM errors in annotation may not be uniform throughout the data. In other words, the LLM may have more error for particular subsets of the data than others, which can cause bias in downstream analysis that uses LLMs as an input. We recommend using a de-biasing method to correct for those errors in downstream analysis (see Egami et al. 2023, 2024).

Because of the challenges of pure reproducibility when using closed-source models, we recommend that researchers take the following steps during validation:

First, researchers should always specify the exact model version (e.g., GPT-5.2.-2025-12-11 not GPT-5.2) as performance can vary meaningfully across releases from the same model family (Barrie, Palmer, and Spirling 2025).

Second, researchers should always revalidate when running new data collection even with similar models. To aid in this, we recommend that researchers set up their workflows to make swapping out model versions straightforward.

Third, as we discuss in the next section, researchers should make their workflows as auditable as possible.

Fourth, researchers should calculate and consider both cost and performance when deciding on models.

A challenge that researchers are beginning to face is what to do when models perform better than humans. Traditionally, we have validated machine performance against a benchmark of humans but, in some use cases, LLMs appear to outperform humans. Briggs et al. (2025) use a reconciliation approach where human RAs and multiple LLMs label the same content. Disagreements between coders of any type are then escalated to an expert human who reconciles the correct answer (while blinded to which answer was given by humans versus LLMs). A dataset of consensus codes and reconciliations is then used as the gold standard.

The situation we have not yet faced is when the LLM might have superior performance even than the reconciler. At that point, it is not clear how LLM performance could be validated. However, it would be likely that the reconciler approach would still indicate that the LLM was outperforming humans, but it would become difficult to tell by how much. We return to this point in Section 10.9.

10.3 LLM-GENERATED TREATMENTS AND STIMULI

10.3.1 The Method

Generating treatments and stimuli for an experiment is often the most unique and customized phase of experimental design and analysis. This is particularly true for designs using complex stimuli like text, images, or audio, where

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the causal variable of interest is embedded within media that often simultaneously vary on many other dimensions. Researchers must think carefully about every aspect of what they show participants: how treatments differ from one another, what participants might infer beyond the intended manipulation, and what associations a given stimulus might evoke. The more complex the stimuli design – be it using videos with many aspects that could change between two treatments or a dynamic interaction – the more difficult it is to ensure that the variable researchers intend to test is, in fact, responsible for observed differences across conditions.

In practice, producing these treatments has traditionally required researchers to create or source all stimuli themselves. While best practices exist for various designs (Aguinis and Bradley 2014; Butler and Crabtree 2021), answering a specific research question still requires considerable discretion, and piloting and refining treatments can be slow and expensive. Some stimuli pose particular logistical challenges: creating images or videos where only a single element varies between conditions may be difficult or impossible; and training confederates for interactive experiments is costly, time-consuming, and difficult to precisely control.

To address these challenges, researchers have begun using LLMs for different forms and phases of treatment design. These include generating entirely novel stimuli (Evsyukova et al. 2025), creating new treatments based on existing examples (Porter and Velez 2022; Goldstein et al. 2024), tailoring treatments to individual respondents (Velez and Liu 2025; Hackenburg and Margetts 2024), and deploying an LLM as a conversational agent that directly interacts with participants (Argyle et al. 2023a; Yin et al. 2024; Argyle et al. 2025b). There is significant potential across these use cases – especially as relates to challenges of scaling treatments and having precise control over differences between treatments – that are difficult for the average researcher to generate, such as images. Yet they also raise new difficulties: validating construct fidelity at scale, navigating the ethics of LLM-participant interactions, and understanding whether automated generation itself affects treatment effects. Large language models expand what researchers can feasibly study, but at the same time they compound many existing challenges of experimental design.

10.3.2 Where We Came From

Generating experimental treatments has often been a creative exercise for researchers. This is particularly true for multidimensional treatments: Text-based treatments often vary in tone, length, and word choice; images commonly have different backgrounds and subject features; and treatments involving interactions with confederates commonly vary in content and style. Designs featuring these sorts of treatments confront a common challenge: isolating the causal dimension(s) of interest while holding others constant in

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expectation. While the implications of this challenge are well-studied (e.g., Dafoe et al. 2018; Damann et al. 2025), the construction of treatments still involves considerable researcher discretion.

Experiments using multidimensional treatments have been particularly productive in the study of media and political communication, where multidimensional treatments are often necessary to effectively represent the media of interest. Text-based treatments, developed in survey experiments, enabled researchers to make strides in substantive areas involving shortform text (e.g., names in audit studies) as well as longer-form text (e.g., news articles). However, these text-based manipulations often change more than the causal variable of interest. For example, in an audit study, changing an applicant's name might also manipulate the reader's assumptions about race, ethnicity, and socioeconomic class (Landgrave and Weller 2022). Similar concerns arise with treatments involving human confederates, rule-based chatbots or bots on social media, and visual stimuli.

Large language models offer many potential benefits to these designs. Two especially promising possibilities emerge. First, by reducing the costs of text generation, LLMs make it possible to generate many more variants of each permutation of manipulated conditions, making it practical to use many treatments per condition rather than relying on a single carefully crafted exemplar. Second, in some cases, they improve the fidelity of the intended manipulation, particularly in designs where treatment is difficult to control, as with human confederates (Argyle et al. 2023a). But at the same time, at minimum, they simply transform the concern about multidimensionality and potentially amplify it. Researchers must now contend with learned associations in model outputs, correlated features of training data that a researcher often cannot observe, and the difficulties associated with auditing a larger volume of stimuli (Bai et al. 2025a). Nonetheless, the potential benefits have motivated rapid adoption, and Section 10.3.3 examines how researchers have applied LLMs to treatment generation.

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10.3.3 What This Method Does

In practice, the basic operation is straightforward: Researchers provide instructions to language models specifying the content of stimuli, often explicitly attempting to hold some features constant and specifying which are meant to vary. However, this appeal rests on two mechanisms. First, LLMs can generate many instances of treatment. In practice, this may make it tenable to assume that idiosyncratic variations in nontarget dimensions of experimental stimuli average out. Second, a language model allows the researcher to instruct the model to explicitly hold certain features constant while varying the treatment along other dimensions. Yet the fundamental problem remains: instructing a model to vary one dimension does not guarantee other dimensions stay constant.

With this concern in mind, what is distinctive about LLM-generated experimental stimuli? Large language models transform the challenge posed by multidimensional treatments but do not eliminate it completely. The concern shifts from manually producing treatments varying only in the target dimension to ensuring that the model's learned associations do not introduce unintended covariation between treatment and potential confounders. Because models are trained on naturalistic data, prompts that are otherwise equal except for the target dimension may not yield outputs that only vary in the target dimension. As the subsequent examples demonstrate, it is still incumbent on researchers to validate experimental stimuli, though these examples also demonstrate the promise and potential benefits of LLMs in these use cases.

10.3.4 Example Applications

10.3.4.1 Generating Static Treatments

The most straightforward application is generating static treatments for randomized experiments. For example, consider audit studies: Scholars have critiqued the traditional approach of signaling race through names, noting that names may simultaneously convey information about ethnicity, socioeconomic status, and other attributes (e.g., Landgrave and Weller (2022), as discussed earlier). Evsyukova et al. (2025) sidestep name-based considerations by creating fictitious LinkedIn profiles with AI-generated images that vary applicant race directly. Profiles send connection requests, and researchers observe acceptance rates. The authors argue this approach addresses privacy concerns associated with using real photos, and crowdworkers could not distinguish generated images from real ones. Beyond privacy, AI-generated images could allow researchers to hold constant other features (e.g., background, clothing, age) and isolate the effect of race more precisely. However, the concern about learned associations applies here as well: Researchers must verify that generated images vary only on the target dimension, since models may introduce race-correlated features (e.g., background settings, attire, or image quality) that the researcher did not intend.

Beyond images, researchers have used LLMs to generate text-based treatments across a variety of research designs. For example, Goldstein et al. (2024) create many novel propaganda items based on existing examples, producing a volume that would be difficult to generate manually while allowing idiosyncratic variation to average out across items. Lai et al. (2026) conduct a conjoint experiment in which LLM-generated social media posts systematically vary anti-elite rhetoric, people-centric appeals, policy topics, and ideological direction. Porter and Velez (2022) use LLMs to create placebo control news stories that function as counterfactuals for articles selected to serve as treatment conditions. Velez and Liu (2025) use an LLM to scale treatments in a different way, generating counter-attitudinal information for each participant

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in response to previous answers they had given in the survey. Similarly, Hackenburg and Margetts (2024) create persuasive messages micro-targeted to participants based on their demographic and political attributes. These applications enable research designs that would previously have been impractical, but they also raise validation concerns.

10.3.4.2 Interacting with Participants

Another promising direction is using LLMs as conversational agents to test theories that rely on interaction. Training human confederates for conversational experiments requires substantial resources and limits sample sizes; LLMs can interact with thousands of participants following consistent instructions. Argyle et al. (2025b) test how conversation length and message targeting affect political persuasion, instructing GPT-4o on the number of exchanges and whether to tailor arguments to individual conversation partners. They find no significant difference between messaging strategies, with statistical power that would be difficult to achieve without an LLM.

In addition to having persuasive conversations, some researchers have tested the abilities of language models to mimic emotional processes in conversation. Yin et al. (2024), for example, test whether conversations with an LLM could make people "feel heard." Other studies use LLMs as an assistant in person-to-person conversations, rather than as a substitute – for example, to moderate extreme language (Argyle et al. 2023a). These applications lower barriers to studying conversational processes, particularly for resource-constrained researchers, but raise concerns about validity, replicability, and ethics.

10.3.4.3 Other Uses

Though less central, there are several other ways in which researchers have begun to use LLMs to facilitate the experiment design process. In addition to generating treatments, some researchers have used LLMs to generate questions for use in experiments. Velez and Liu (2025) do so within an experimental context, developing specific questions for each participant to assess their reactions to the information which was generated for them based on their preexisting attitudes. Alternately, Götz et al. (2024) do so as a preparatory step, using GPT-2 to generate many potential scale questions, based on several starting questions, to measure a particular construct, which can then be validated as with any question battery as part of scale creation.

Finally, some researchers have proposed using LLMs not to generate experimental materials but to validate them. Ashokkumar et al. (2026) develop tools that estimate treatment effects and statistical power from proposed treatments and survey questions, which would typically require pilot studies or guesswork. However, such estimation likely works best for treatments and questions similar to those used in prior research. Estimates about novel stimuli that fall outside the model's training data would likely be inaccurate.

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10.3.5 Validation and Inference

Though LLMs offer unique opportunities for generating experimental materials, their use compounds existing challenges associated with writing stimuli. The first is construct validity. When designing treatments manually, researchers must ensure that the intended causal dimension is what changes and that participants observe only this manipulation. Validating LLM-generated stimuli follows a similar logic but is complicated by the very features that make LLMs attractive: scale and complexity. When generating many treatments, at what point can researchers be confident that successive outputs accurately represent the target construct? When LLMs interact with participants, how many input permutations must be tested before researchers trust that all participants receive valid treatment?

A related challenge is minimizing unintended covariation – what is sometimes called "bundled" treatments – where multiple dimensions vary simultaneously and their effects cannot be separated. Isolating the target dimension while holding nontarget dimensions constant is difficult with any multidimensional treatment; LLMs introduce two additional complications. First, as noted previously, validation becomes harder at scale. Ensuring that treatments vary only on the target dimension is simpler for single sentences than for paragraphs, conversations, images, or videos, yet LLMs make generating these more complex stimuli easier. Second, as with many phases of research design, the process of designing treatments or training human confederates often forces the researcher to clarify theoretical expectations. Crafting treatments by hand forces researchers to consider which elements matter, what mechanisms might operate, and what unintended cues participants might detect. Large language model generation may shortcut this process, potentially to the study's detriment. Finally, LLM-generated treatments introduce a novel concern: disclosure effects.

Several studies show that when participants know they are interacting with an LLM, treatment effects are attenuated (Yin et al. 2024; Palmer and Spirling 2023). This raises questions about whether findings generalize to non-LLM contexts. Additionally, unmonitored LLM interactions with participants pose ethical concerns, particularly for longer or more sensitive conversations.

Given these challenges, extensive validation must be central to any use of LLMs for treatment generation. When generating a relatively small number of stimuli, researchers should identify the exact elements that they believe are responsible for any observed effects and justify other variation between elements. For larger scale treatment generation, all LLM output should be recorded and released as a component of the final data. Such output should itself be analyzed as media to assess feature variance across treatments, variance within treatments, and how this correlates with final outcomes. Finally, where possible, unless LLMs are the subject of the study, LLM-generated treatments should be compared to human-generated treatments, even if on a relatively small scale.

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10.4 SILICON SAMPLING

10.4.1 The Method

Silicon sampling, also known as digital twins (Toubia et al. 2025; Peng et al. 2025), is a research methodology in which LLMs are used to generate synthetic proxies or replicas of individual-level research data (Argyle et al. 2023b; Kozlowski and Evans 2025; Chapter 3). The basic approach involves prompting (or otherwise training or tuning) a language model to provide a response that approximates some unknown quantity on the basis of provided information that constructs a particular "persona." For example, the model might be provided with party identification, race, gender, and other demographic or attitudinal characteristics and asked to provide the expected vote choice for that constructed individual. The provided persona constructions vary in complexity, and the generated data can be used to proxy specific questions, full surveys, semi-structured interviews, or responses to experimental stimuli. The purpose is to generate lower cost alternatives to inform, augment, or replace human subject data collection in behavioral science research.

Silicon sampling is controversial on both practical and theoretical grounds. On the practical front, there is significant debate about the extent to which political and social phenomena can be fully and completely represented by LLMs. Existing research provides demonstrations that are both more optimistic (Argyle et al. 2023b; Dillion et al. 2023; Horton 2023; Goli and Singh 2024; Ashokkumar et al. 2026) and more pessimistic (Peng et al. 2025; Santurkar et al. 2023; Boelaert et al. 2025; Bisbee et al. 2024; Wang et al. 2025a; Qu and Wang 2024).[2]

However, proponents cite the lower cost (Broska et al. 2025), instantaneous data, and potential for exploration of questions that may be challenging or unethical to explore with human subjects. This is even more poignant as widespread LLM use by online panelists leads to questions about data quality in existing sources and the likelihood of rising costs for human survey data in the future (Westwood 2025; Veselovsky et al. 2025). Both start-up companies and established survey panel providers are promoting products for marketing insights using completely in silico data.

On theoretical grounds, there is also an important concern about the validity of replacing humans in research on human attitudes, psychology, and behaviors. This reflects concerns about how the fundamental nature of the scientific study of people and communities is distorted when researchers no longer directly observe or interact with their human subjects. Additionally, even if scholars can demonstrate that silicon sampling methodologies provide good replications in a particular application, it remains unclear what standards

[2] See also Section 10.3.4.

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would provide enough confidence in synthetic data to use it for novel scientific inference without validation against a human sample.[3]

10.4.2 Where We Came From

Different groups of scholars bring different intellectual traditions to their assumptions and approaches regarding silicon sampling. There are at least three major traditions in this work.

First, some authors approach silicon sampling from a perspective of deriving insights from second-hand uses of "found" big data (see Grimmer 2015). Second, other authors approach the data from a tradition of unrepresentative samples and online crowd work (Porter et al. 2020; Kittur et al. 2013). Finally, still other scholars bring a lens of agent-based modeling. The first two traditions are briefly reviewed here. Readers are directed to Section 10.5 for a more complete discussion of agent-based modeling and LLM simulation.

Large language models are trained on massive swaths of written human text (and multi-modal audio or visual inputs for some models). Thus, some scholars approach LLM simulation from the conceptual background of machine-learning algorithms and "big data." Because human authors produced (at least a nontrivial amount of) the training material and because the LLM learns many patterns of meaning and conceptual association that go far beyond just rote encoding of rules of grammar (Wei et al. 2022), some of the associations encoded into the LLM weights reflect human patterns of thinking, behaving, speaking, and valuing (Kozlowski and Evans 2025). Unwanted and undesirable emergent patterns are termed "algorithmic bias" (Bolukbasi et al. 2015). Large language model–generated content reproduces known patterns of partisan, racial, gender, national, or other biases (Gallegos et al. 2024; Chapter 5; Chapter 6). However, while normatively undesirable in society, valuable social scientific patterns, termed "algorithmic fidelity" by Argyle et al. (2023b), may be encoded alongside the biased expressions of human psychology and behavior. Scholars in this approach may view interacting with the LLM as an exercise in navigating the black box to effectively surface the right associations from the trillions of potential parameters in a model. Thus, like many big data applications, although the whole body of an LLM's output may be biased and unrepresentative (Santurkar et al. 2023; Wang et al. 2025a), scholars may be able to carefully construct prompts and interactions that elicit scientifically meaningful representations of populations and behavioral processes.

Another set of scholars comes to silicon sampling from a perspective of crowdwork and adapting to available but unrepresentative – and sometimes unreliable – respondent populations. Large language models are agents trained to do particular tasks, but they may not naively be representative of whole populations. Prompts, fine-tuning, and other methods of instructing the LLM

[3] See Chapter 3 for much more in this regard.

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provide guidelines for LLM behavior which, when coupled with their massive training backgrounds and emergent abilities, enable them to engage in a wide range of tasks including those related to silicon sampling. While not specifically conducting silicon sampling, Vafa et al. (2025) conceptualize the importance of LLM steerability – meaning that models provide appropriately varied output based on different prompts – and propose a novel metric for evaluating model steerability. If models are appropriately steerable, then they can provide a range of different attitudes that may be sufficient to reflect a whole population. However, it is not clear what the sampling frame is for an LLM or how to think about randomness or representativeness in sampling from that black box. Authors in this space view statistical calibration of results as a necessary, but likely not sufficient, research step to improve the validity and reliability of output (Vafa et al. 2025; Hullman et al. 2025; Egami et al. 2023; Imai and Nakamura 2025; Broska et al. 2025).

10.4.3 What This Method Does

Silicon sampling is modeled on survey research, and the end result is typically an expected individual-level dataset but with a radically different data-generating process. At their core, LLMs are next-word prediction engines. Given a prompt with certain characteristics, the LLM is expected to respond to further survey questions in a way that is likely, given the "persona" it received as an input. Because different words in a prompt produce different probabilities for the next token, a single LLM can be used to produce a wide variety of outputs that generate expected patterns of variation based on the prompts.

The most basic approach to silicon sampling sets up a template for a persona (often using a "system" prompt to create an overarching set of instructions), which pipes in standard fields for a variety of descriptive characteristics of a given persona and asks the LLM to produce an additional piece of information. In some cases, the personas represent a complete set of possible profiles given the possible intersections of the various background information, with each persona-prompt being run multiple times to create a distribution of responses. In other cases, researchers benchmark an existing survey data source, using actual demographic profiles of respondents to surveys, such as the American National Election Study or World Values Survey, to generate persona prompts that are "digital twins" of human subjects.

Although this sounds straightforward, there are several essential decisions that can have important implications for the survey results. While these are by no means the only relevant decisions, two crucial researcher choices are: the information contained in the prompt and model selection.

First, what information should go into the prompt? When human subjects answer survey questions, survey researchers assume the subject exists as a unique and independent entity and the purpose of survey questions is to objectively observe and measure various aspects of that person as they really exist.

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There are well-known exceptions to this assumption in survey research, such as a "Hawthorne effect" in which the mere act of being observed changes a behavior. Social desirability bias, interviewer effects, and other survey design and question wording decisions affect whether it is possible to observe the survey taker without influencing the construction of the responses. But most survey research practices aim to minimize the impact of these researcher interferences (Weisberg 2009).

In silicon sampling, however, the persona is a digital representation of a real or hypothetical person who is constructed entirely through the training and prompting of the model. Thus, information that is not directly given to the model in the prompt cannot be assumed to be known or a part of that person's identity, in the same way that questions not asked by the researcher can still be assumed to be constitutive of the human. For example, the gender of a human survey respondent would be assumed to have an impact on their worldview even in a survey that does not record the respondent's gender. But if gender is not provided in the prompt to an LLM, the persona does not have a defined gender and the LLM will use an unknowable process of probability calculation to account for that missing information (see discussion in Gui and Toubia 2023). That is all to say: giving additional and more relevant information to the LLM is likely going to improve prediction performance when compared with giving less information. However, it might also lead to the model placing outsized importance on enumerated background characteristics or result in the information provided being so comprehensive as to render the model contribution trite.

10.4.4 Example Applications

There are two important ways that researchers have pushed the methodology forward in giving more information to the LLM for the purposes of silicon sampling. Some authors provide backstory information that is substantially more detailed (Park et al. 2024; Toubia et al. 2025). Park et al. (2024) conducted multi-hour in depth interviews with 1,000 respondents to incorporate a rich history into silicon sampling generations that is not typically available in a survey. Using this information in the prompts resulted in high performance on prediction of later survey questions.

Another approach to providing more information to the model is moving beyond using only the prompt to generate the persona and, instead, directly fine-tuning the model (Binz et al. 2025; Kim and Lee 2023; Suh et al. 2025). Fine-tuning is a process of using an end-stage training data set to adjust the model weights to incentivize behavior that more closely matches a specific, desired outcome set for a particular task. Fine-tuning requires higher technological capacity and computing resources, it may unexpectedly not only affect other model functions (Luo et al. 2025) but can improve performance enough to be worth the investment in some cases.

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Second, in addition to determining the process and information, the model will use to construct the persona, the researcher must also select a model to use. One primary decision is whether to use an open-weight model or a closed-weight model (see more discussion of the trade-offs in Section 10.4.5). However, even beyond this high-level decision, there is evidence that differences in training across model families – and post-training steps within model families – can have important, direct effects on model performance for silicon sampling tasks (Lyman et al. 2025; Boelaert et al. 2025). For example, models of different sizes, in different families, or with different post-training may have different biases in the response distributions, different capacity to complete the task, or different refusal triggers. One important implication of this heterogeneity is that prompting should be done in a model-specific way, and researchers may need to craft different prompts to get comparable performance from different models, as different models may interpret an identically worded prompt to be suggesting a qualitatively different kind of task. This is consistent with research demonstrating that even relatively minor differences in prompt structure, such as wording or ordering of otherwise identical information, can have a substantial impact on silicon sampling output (Brucks and Toubia 2025).

10.4.5 Validation and Inference

Existing empirical demonstrations of silicon sampling come to many competing conclusions about the potential success, failure, and biases or limitations of the approach. In addition to variation that is introduced by model selection and prompting strategies, nascent research in this space has not yet coalesced around an appropriate set of metrics. Existing validation approaches include individual-level prediction accuracy metrics (F1 scores, percent matches, or correlations between human and silicon observations), distributional characteristics (means, proportions, or variances), or relational statistics (subgroup analyses, treatment effects, or coefficient estimates between independent and dependent variables). Kozlowski and Evans (2025) provide a useful framework of potential pitfalls and associated validation practices for silicon sampling.

Choices of different metrics for validating silicon sampling are signals of fundamentally different inferential goals across studies. Thus, determining which is the appropriate metric depends on clearly identifying the kind of inference that scholars want to make using the synthetic data (Argyle et al. 2025a; Chapter 3). For demonstration purposes, Sections 10.4.5.1–10.4.5.3 briefly describe three examples of different underlying assumptions about the inferential purpose of a silicon sampling application that imply different validation metrics.

10.4.5.1 Individual-Level Imputation

As noted in Section 10.3, Kim and Lee (2023) present a fine-tuning-based approach to silicon sampling for data imputation tasks. In an imputation task,

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the silicon data will be used alongside human data to make statistical inferences about the attitudes or behaviors of specific survey respondents. Thus, the goal is to exactly predict the response that a specific given respondent would have had to the question at hand, had they provided it at a given point in time. In this case, the authors choose to use metrics that test prediction accuracy by directly comparing a specific human response to the probability that the model selected that response using an AUC (Area Under the receiver operating characteristic Curve). Other scientists use more standard metrics of proportion agreement or Cohen's kappa (Qu and Wang 2024).

10.4.5.2 Simulations of Aggregated Distributions

In other applications, the goal is not necessarily to predict a single individual precisely but rather to replicate the distribution of opinions in a given population or subpopulation. Many early attempts at silicon sampling compared measures of central tendency, such as proportions or means, from synthetic data to human data finding clear signals (Argyle et al. 2023b). However, later scholars emphasized different metrics of variation or distributional overlap as important factors for inferences based on distribution, finding that silicon samples often lacked the required variation. Specifically, Bisbee et al. (2024) note that, if researchers want to use silicon data for pretesting or power analysis, then variance is an essential element for valid inferences. Variation in responses can be dramatically affected by both temperature (which is subject to researcher control) and post-training alignment process (typically outside of researcher control) (Kirk et al. 2023), making model selection and evaluation essential when prioritizing distributional variation.

10.4.5.3 Simulations of Relationships

Still other researchers want to use silicon sampling methodology in a causal inference toolkit to investigate potential relationships or treatment effects (Binz et al. 2025; Imai and Nakamura 2025; Gui and Toubia 2023). From an observational perspective, one approach is to use subgroup analysis or estimated coefficients to carefully investigate whether silicon samples are able to produce expected relationships between key independent and dependent variables (Santurkar et al. 2023; Argyle et al. 2023b). Typically, that approach finds that silicon samples are more accurate for majority groups or Western countries (Wang et al. 2025a; Santurkar et al. 2023; Qu and Wang 2024). From an experimental perspective, scholars have evaluated whether silicon samples can reproduce treatment effect estimates (Gui and Toubia 2023; Aher et al. 2023; Ashokkumar et al. 2026; Horton 2023), perhaps with additional statistical corrections (Broska et al. 2025; Hullman et al. 2025). If such effects could be reliably estimated, then LLMs might provide useful tools for piloting experiments or selecting treatment materials. While many (but not all) silicon experiment replications produce expected directional effects, effect size can be incorrect (Cui et al. 2025). Furthermore, the presence of previously published

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experiments in training data raises questions about the validity of the simulation, known as data leakage (but see Ashokkumar et al. 2026 who find similar performance simulating published and unpublished experiments).

Other researchers may bring additional motivations to silicon sampling approaches, which may imply still other appropriate qualitative evaluation or statistical tests. While best practices and community standards are still being developed, researchers are encouraged to be explicit about the inferential purpose of their silicon sampling approaches and to explicitly state and justify the selection of particular evaluation standards that support the desired inferences (Argyle et al. 2025a).

10.5 AGENT-BASED MODELS AND GENERATIVE AI

10.5.1 The Method

Agent-based models (ABMs) are computational simulations composed of multiple autonomous entities ("agents") that perceive their environment, maintain internal states, and select actions. The methodology proceeds by specifying (1) a population of heterogeneous agents; (2) interaction structures such as networks, spatial environments, or institutions; and (3) behavioral rules and update schedules. Researchers then observe the macro-level patterns that emerge from repeated local interactions (Bonabeau 2002; Macy and Willer 2002; de Marchi and Page 2014).

The core theoretical commitment is generative: ABMs seek to grow collective phenomena from plausible micro-foundations, rather than stipulating them and learning from the distribution of outcomes across many simulation runs (Epstein and Axtell 1996; Axelrod 1997). This contrasts with approaches that treat collective outcomes as simple aggregations of individual attributes, emphasizing instead how macro-level patterns arise from micro-level interactions.

Generative ABMs (GABMs) – sometimes called "agentic" or "multi-LLM" systems – replace hand-coded behavioral rules with LLMs. Instead of explicit heuristics or equations governing agent choice, an LLM serves as a context-sensitive policy function: Given a role, goals, memory, and institutional constraints, it generates the agent's next action (typically communicative) and updates its internal state (Park et al. 2023; Lu et al. 2024).

Crucially, this involves not just an LLM but an entire architecture: Prompts encode roles and constraints, memory modules compress experience, orchestrators manage turn-taking and tool access, and evaluators assess outputs (Park et al. 2023; Wu et al. 2023b; Zhou et al. 2024). This approach enables new forms of expressiveness: Language, persuasion, justification, and norm enforcement become endogenous simulation dynamics rather than exogenous labels or post-hoc interpretations. That is, political concepts – ideology, framing, persuasiveness – can be operationalized through generated linguistic behavior (and then measured), rather than only as fixed scalar attributes.

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10.5.2 Where We Came From

Schelling's segregation model (Schelling 1971) remains the canonical example of agent-based modeling in social science because it demonstrates the power of interaction-driven emergence. Schelling showed that extreme residential segregation can arise even when individuals hold only mild preferences for same-group neighbors – a macro-pattern that cannot be predicted from simply aggregating individual tolerance levels. This demonstration of how simple micro-rules generate complex macro-outcomes catalyzed broader adoption of ABMs across the social sciences.

By the 1990s and early 2000s, ABMs had established a substantial lineage in political science, sociology, and economics (de Marchi and Page 2014; Macy and Willer 2002). More recently, this tradition has cross-fertilized with computational social science approaches that leverage large-scale digital trace data as input for formalized models of network dynamics and information flows (Lazer and Ognyanova 2024; Lazer et al. 2020).

Since the consolidation of ABMs as a generative methodology, a series of influential contributions demonstrated how simulation can link micro-level rules and interaction structures to macro-level regularities. Epstein and Axtell (1996) offered a paradigmatic demonstration of "growing" societal-level patterns from simple behavioral and environmental primitives, while Axelrod (1997) showed how adaptive agents engaged in repeated interaction can generate robust patterns of cooperation and norm-like behavior. Work in a similar spirit used computational models to account for specific empirical regularities: de Marchi (1999) formalized how adaptive dynamics and feedback can produce electoral instability, and Cederman (2003) connected micro-level interaction mechanisms to macro-level distributions in war severity.

As we will see further on, contemporary generative ABMs exhibit a comparable breadth of ambition, ranging from society-scale simulations of interacting populations to narrowly targeted models designed to probe why particular outcomes emerge under specified institutional and informational constraints.

Canonical political-economy ABMs specify four components: (1) a heterogeneous population of agents; (2) an environment (spatial, network, or institutional) governing interaction patterns; (3) micro-rules mapping local observations to actions – typically boundedly rational heuristics rather than full optimization; and (4) state-transition functions updating agents and environments after each interaction. Whether agents play iterated strategies in repeated games (Axelrod 1997) or follow simple behavioral rules on spatial grids (Epstein and Axtell 1996), the research logic is identical: characterize how macro-outcomes vary with micro-rules, interaction structures, and initial conditions across many simulation runs.

The traditional ABM approaches offered political scientists three advantages: interpretability by construction (explicit behavioral rules made mechanisms traceable), computable institutions (formal rules became manipulable

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parameters for counterfactual analysis), and experimental control (systematic variation of micro-assumptions revealed their distributional consequences) (Epstein and Axtell 1996; Axelrod 1997; de Marchi and Page 2014).

These methods nonetheless struggled with behavioral realism: Compact decision rules could not capture linguistic and interpretive political behaviors like persuasion, framing, or norm contestation, which were reduced to scalar updates or thin heuristics. Measurement alignment also proved difficult: Internal states rarely mapped cleanly to observables, complicating validation. And because ABMs prioritize mechanistic explanation over prediction, external validity claims remained limited to "mechanism X can generate pattern Y" without strong forecasting power (de Marchi and Page 2014).

As we will see later, GABMs are able to overcome some of the problems with behavioral realism. Specifically, we can use these technologies to populate a system with agents that are endowed with the ability to interact with language rather than some pre-specified distributional formalism. In so doing, however, we also introduce problems relating to mechanistic interpretability and replicability. Overall, the "generative" shift with LLMs should be read as changing (1) the representation of agent cognition and discourse and (2) the practical workflow for building and iterating environments. It does not replace core ABM standards: Explicit mechanisms, controlled experiments, and transparent documentation remain the baseline.

10.5.3 What This Method Does

So, how are practitioners using generative AI for agent-based modeling? Pseudo-code 1 and 2 describe the higher-level attributes of the traditional ABM and more recent GABM approaches (see Figure 10.1). In the ABM panel, heterogeneous agent attributes parameterize explicit decision rules that map observations to actions, after which interactions generate outcomes and state transitions that can be aggregated across runs. In the GABM panel, personas and memory retrieval form part of the agent's context for a black-box LLM policy that emits both natural-language output and structured actions, with transcripts and logs becoming additional artifacts for measurement and replication.

This points to three components of new GABM approaches that distinguish these methods from prior agent-based modeling. First, instead of representing heterogeneity primarily as numeric attributes or fixed types, GABMs often define agents through personas: natural language profiles that encode identities, priors, goals, and situational constraints. Personas can be coarse (e.g., membership in the red team versus the blue team) or richly parameterized from survey microdata and other sources, allowing researchers to inject structured variation into populations while still enabling context-sensitive linguistic behavior (Park et al. 2023; Törnberg et al. 2023).

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Algorithm 1 Toy ABM (hand-coded rules)

Inputs: agent attributes (θ_A, θ_B); initial states (x_A, x_B); environment S_0
for $t = 1$ to T **do**
  $o_A, o_B \leftarrow \text{OBSERVE}(S_{t-1}, x_A, x_B)$
  $a_A \leftarrow f_A(o_A; \theta_A)$; $a_B \leftarrow f_B(o_B; \theta_B)$
  $yt \leftarrow g(a_A, a_B, S_{t-1})$ ▷ micro outcome
  $(x_A, x_B, S_t) \leftarrow h(x_A, x_B, S_{t-1}, yt)$ ▷ state transition
end for
$\widehat{M} \leftarrow \text{AGGREGATE}(\{yt\}_{t=1}^{T})$ ▷ macro outcome

Algorithm 2 Toy GABM (LLM agents)

Inputs: *personas* (p_A, p_B); goals; initial states; environment S_0; memories M_A, M_B
for $t = 1$ to T **do**
  $m_A \leftarrow \text{RETRIEVE}(M_A, S_{t-1})$; $m_B \leftarrow \text{RETRIEVE}(M_B, S_{t-1})$
  $a_A, \text{text}_A \leftarrow \text{LLM}(p_A, o_A, m_A)$ ▷ black-box policy + language
  $a_B, \text{text}_B \leftarrow \text{LLM}(p_B, o_B, m_B)$
  $M_A \leftarrow M_A \cup \{S_{t-1}, \text{text}_A, \text{a}_\text{A}\}$; $M_B \leftarrow \cdots$
  $yt \leftarrow g(a_A, a_B, S_{t-1})$
  $(S_t) \leftarrow h(S_{t-1}, yt)$ ▷ state transition
end for
$\widehat{M} \leftarrow \text{AGGREGATE}(\{yt\})$ and / or $\text{MEASURE}(\{\text{text}_t\})$

FIGURE 10.1 Pseudo-code: traditional agent-based model and generative agent-based model architectures.

Second, many GABMs implement persistent memory as retrieval over prior experiences, observations, and interactions. Whereas classic ABMs typically store state as a small vector (e.g., beliefs, payoffs, or strategy parameters), generative agents store compressed natural language records of events and then selectively retrieve them when acting. This makes it easier to represent path dependence in communication and interpretation (e.g., reputations, grudges, learned narratives, agenda carryover), but it also introduces new modeling choices that must be specified and logged (what gets stored, how it is summarized, and what is retrieved at each step).

Third, GABMs replace the hand-coded decision rule with an LLM policy: a stochastic, black-box mapping from context to actions (and often to accompanying language). In practice, the agent's "policy" is implemented as a prompt bundle (system rules + persona + local observations + retrieved memory) passed to an LLM whose outputs depend on sampling and decoding settings (e.g., temperature, maximum tokens). Because unconstrained generation can drift or violate required formats, GABMs often add guardrails: constrained action schemas, validators/parsers, retry and fallback logic, and sometimes an adjudicator that enforces institutional rules. Methodologically, these decoding and control parameters are part of the model specification, since they shape variance, path dependence, and the stability of multi-step trajectories.

A fourth component, often implicit in traditional ABMs but explicit in GABMs, is the orchestration and evaluation components. Orchestration refers

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to the controller that schedules interactions (who speaks/acts and when), gates observations (what an agent is allowed to see), assembles prompt context (personas, retrieved memory, institutional constraints), and routes tool access and guardrails. Evaluation refers to the measurement layer that audits agent behavior and converts language traces into analyzable outcomes (e.g., by classifying text into states or scoring toxicity/civility), typically alongside ablations and robustness checks. These components are visible in the applied examples: Park et al. (2023) rely on an environment server and planning/memory loops plus ablation-based assessment; Chuang et al. (2024) enforce a structured interaction schedule and map tweets into discrete opinion states via classification; and Törnberg et al. (2023) treat the feed-ranking rule as the orchestrated institutional mechanism and operationalize outcomes from generated text under controlled counterfactuals.

Together, these components change what ABMs can represent. Classic ABMs often reduced political interaction to scalar belief updates or stylized heuristics because rich interpretation and language were expensive to model explicitly. By contrast, GABMs make it comparatively cheap to model political behavior that is mediated by text: persuasion and counter-persuasion, framing contests, threat and reassurance signaling, norm contestation, and so on. The inferential logic remains ABM-like – macro-patterns are still functions of micro-assumptions and interaction structures. But the micro-foundations can be substantially richer in settings where meaning-making and communication are central and where researchers may wish to treat the evolving content of interaction as both a causal channel and an empirical object of measurement.

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10.5.4 Example Applications

Emerging GABM applications can be organized by simulation scope: (1) open-ended "whole society" environments generating rich interaction traces; (2) macro-phenomena (polarization, consensus) as emergent outcomes of language-mediated network interaction; and (3) targeted institutional interventions comparing counterfactual rule designs.

10.5.4.1 Whole-Society Simulation

Park et al. (2023) populate an artificial town with twenty-five LLM agents, each maintaining memory streams, retrieving relevant experiences, and periodically synthesizing higher-level inferences. The environment constrains observation (locality) and interaction (turn-taking), while language traces (conversations, plans) are logged as analyzable artifacts. Their canonical demonstration – a party organized through conversational diffusion without scripted coordination – illustrates how social organization can emerge from transparent micro-rules. Methodologically, this establishes a recipe for open-ended emergence with inspectable instrumentation.

10.5.4.2 Macro-level Opinion Dynamics

Chuang et al. (2024) adapt classical opinion-dynamics ABMs to language-mediated interaction: Agents exchange tweets conditioned on personas, update beliefs through verbal responses, and produce discrete opinion states via classification. This design recovers familiar ABM metrics (trajectories, fragmentation) while retaining linguistic traces of framing and persuasion. Critically, they diagnose that default LLM behavior converges toward consensus unrealistically, then show how prompt engineering and memory design can reintroduce confirmation bias to match empirical polarization patterns. This provides a template for macro-phenomena GABMs with expressive agents and comparable metrics.

10.5.4.3 Institutional Counterfactuals

Törnberg et al. (2023) simulate Twitter-like platforms with agents initialized from survey microdata (American National Election Survey demographics, attitudes, usage). The intervention varies ranking algorithms (engagement-ranked within-network, global engagement, cross-partisan bridging), holding all else constant. Agents post, share, and engage at calibrated rates, with feedback loops between content and visibility. The GABM contribution is operationalizing outcomes directly from generated text (toxicity, cross-partisan engagement) while retaining full transcripts for qualitative audit – a computational sandbox for institutional design.

10.5.4.3.1 ADDITIONAL APPLICATIONS Related work includes society simulators scaled to thousands of agents for policy interventions (Piao et al. 2025; Wang et al. 2025b), election models simulating vote choice under campaign inputs (Zhou et al. 2025), and collective-action cascades with mixed LLM and simple agents (Mou et al. 2024). Reusable platforms are emerging to standardize environments and evaluation (Tang et al. 2025; Zhou et al. 2024; Wu et al. 2023b).

More critical work emphasizes the need for empirical benchmarking and transparency around proprietary models (Palmer et al. 2024; Bail 2024; Barrie and Törnberg 2025; Münker et al. 2025) and raises doubts about the ability of LLMs to faithfully recreate human-like discourse (Pagan et al. 2025). It is to these and related questions we will turn in the final two parts of this chapter section.

10.5.5 Validation and Inference

Generative ABMs inherit traditional ABM challenges – stochastic outcomes, path dependence, and specification sensitivity – while adding new ones: Micro-rules are partly black-box language models where small prompt or decoding changes can produce qualitatively different trajectories (de Marchi and Page 2014; Larooij and Törnberg 2026). Claims should therefore be distributional and conditional: about mechanisms and outcome patterns given an explicit

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design (personas, memory, schedule, model version, sampling parameters), not about single-run narratives.

Following established ABM methodology (de Marchi and Page 2014; Windrum et al. 2007), we propose separating verification (does the system behave as specified?) from validation (does it capture the target phenomenon for its stated purpose?). We provided a bulleted list in Sections 10.5.5.1 and 10.5.5.2 of exactly what we mean by this.

10.5.5.1 Verification

Treat GABMs as experimental systems requiring robustness checks:

- *Sensitivity*: vary prompts, temperature/top-p, interaction order, topology, memory settings.
- *Ablations*: remove memory, reflection, tools, or roles to identify active components.
- *Cross-model robustness*: test whether core patterns persist across model families or versions.

10.5.5.2 Validation

When claiming realism or policy relevance, specify validation targets and calibration strategies:

- *Stylized* facts: match known regularities (fragmentation, homophily, engagement distributions).
- Empirical benchmarks: compare outputs to survey marginals, network metrics, or text corpora when representing specific populations (Larooij and Törnberg 2026).
- Contamination checks: test whether emergent patterns reflect training-data leakage rather than genuine emergence (Ashery et al. 2025; Barrie and Törnberg 2025).
- No circular evaluation: avoid LLMs grading their own outputs as primary validation (Larooij and Törnberg 2026).

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10.6 TOOL USE AND WEB-BROWSING: LLMS AS RESEARCH AGENTS

10.6.1 The Method

Large language model–driven tool use describes a set of research workflows where LLMs are connected to external tools, such as web search, browsing, file readers, and code execution, to help researchers find, collect, and organize data. In this setup, the LLM is not primarily doing the final analysis. Instead, it acts as a flexible interface that takes a researcher's instructions and turns them into concrete steps: searching for sources, navigating websites, extracting relevant information, and assembling the results into structured datasets.

One major advantage of this approach is that it makes it much easier to achieve labor-intensive tasks, such as compiling large datasets, that would normally require a great deal of manual work. For example, a task like collecting biographical information for every member of a legislature, such as the university they attended or their town of birth, would traditionally involve many hours of web searches, copy-pasting, and error checking. With LLM-driven tool use, the same task can often be completed in minutes by combining search, browsing, and extraction tools in a single workflow (Mao et al. 2023; Nakano et al. 2021). This substantially lowers the fixed cost of general information collection and the building of custom datasets, as well as making it easier to iterate and adjust as research questions evolve.

Another important feature of this method is its ability to work with non-text or semi-structured sources. Contemporary LLM systems can extract information from PDFs, scanned documents, tables embedded in images, and other formats that are difficult to handle with traditional scripts (Xu et al. 2020; Kim et al. 2021; Huang et al. 2022; OpenAI 2023). For social scientists, this opens up new kinds of data, such as archival records or administrative documents that were previously too time-consuming to process at scale. In addition, LLMs increasingly function as coding assistants, helping researchers write and adapt scripts for scraping websites or collecting data automatically (Ahluwalia and Wani 2024). This reduces both the time and technical expertise required to build research pipelines, making these methods accessible to a wider range of scholars.

10.6.2 Where We Came From

Historically, the kinds of tasks now handled by LLM-driven tool use were carried out almost entirely by hand. Data collection in the social sciences typically involved researchers or hired research assistants searching for information online (or in person), downloading (or xeroxing) documents, and manually entering values into spreadsheets or databases. Projects that required assembling attributes across hundreds or thousands of cases often took weeks or months and required careful supervision to ensure consistency. As a result, due to temporal or financial constraints, the scale of data collection frequently constrained the kinds of research questions scholars were willing to pursue.

Attempts to automate these workflows existed, but they usually depended on custom scripts written for very specific tasks. Web scrapers or document parsers could speed things up, but they required substantial programming expertise and were often fragile. Small changes to a website's structure or unexpected variation in document formats could cause an entire pipeline to fail. This meant that automation was unevenly adopted and often concentrated among researchers with strong technical backgrounds or access to specialized collaborators.

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Both manual and scripted approaches also raised concerns about accuracy and reproducibility. Human data collection is prone to fatigue, inconsistent judgment, and transcription errors, while traditional automated scripts lack the flexibility to deal with messy or unstructured inputs. In practice, many decisions about how data were collected or interpreted were informal and poorly documented, making it difficult for others to assess or replicate the process. Use of an LLM-driven tool emerges against this backdrop as a way to reduce these long-standing frictions, not by eliminating human judgment but by making data collection faster, more flexible, and easier to document.

10.6.3 What This Method Does

At a practical level, LLM-driven tool use helps researchers offload many of the procedural steps involved in assembling datasets. Instead of specifying every action in advance, researchers can describe their goal at a higher level and allow the model to carry out the intermediate steps, such as finding relevant sources, extracting information, and organizing outputs into structured formats. This makes it easier to move from a research idea to a usable dataset quickly.

A key strength of this approach is its ability to work across different sources and formats within a single workflow. Large language model–driven pipelines can combine web search, browsing, document parsing, and data cleaning that allow researchers to bring together information that would otherwise require multiple separate processes. This is especially useful when working with semi-structured materials like PDFs or scanned documents, which often fall between the cracks of traditional automation tools.

Another advantage is flexibility. Because the model can inspect intermediate outputs and adjust its actions, these workflows are better able to handle real-world messiness, such as inconsistent formatting or missing information. That said, the method primarily operates upstream of analysis. It changes how data are collected and prepared, not how substantive claims are ultimately made.

10.6.4 Example Applications

In practice, LLM-driven tool use is particularly useful for building structured datasets from information that is publicly available but scattered across many sources. In political science, this often includes biographical or institutional data about political elites, which may be distributed across official websites, news articles, and archival records. Tasks like compiling educational backgrounds or career histories, which traditionally required extensive manual effort, can now be completed much more quickly. As a concrete example for illustrative purposes, a modern chat-based LLM was able to collect the alma mater of all sitting US Senators and arrange this information in an extractable table format in a matter of seconds.

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Another common application involves collecting policy documents and administrative materials. Governments and international organizations publish large volumes of reports and regulations in heterogeneous formats, often without consistent metadata. Large language model–driven workflows can automate the retrieval of these documents, extract relevant content, and standardize outputs across cases, making comparative analysis more feasible. As a further example, a modern LLM was able to extract the names of speakers and speaking times from a test set of congressional hearing transcripts and assemble the information in a standardized format, again in a matter of seconds.

These tools are also well suited for monitoring dynamic information environments, such as tracking changes to party platforms or official communications over time. Where previously recollection required restarting a time-consuming human collection cycle, a simple prompt or script can be rerun on a regular basis to ensure up-to-date and consistent information over time. Across these examples, the main benefit is not that LLMs generate new data but that they dramatically lower the cost of assembling data that already exist.

10.6.5 Validation and Inference

Using LLMs for data collection introduces new validation challenges. Because LLM-driven workflows can adapt at runtime, their behavior may vary across executions, even when instructions are held constant. Models may misidentify sources or extract incorrect information, especially in unfamiliar contexts.

To address this, researchers should pay close attention to documentation and provenance. Wherever possible, workflows should retain source links, timestamps, and raw materials, as well as logs of intermediate steps like search queries. While LLM-driven pipelines make this kind of logging feasible, it does not happen automatically and requires deliberate design.

From an inferential standpoint, it is important to separate improvements in data collection from claims about causality or measurement validity. Large language model–driven tool use makes it easier to observe and catalog information, but it does not eliminate the need to think carefully about what the resulting data actually measure. Researchers should therefore be explicit about the limits of the data and cautious in extending claims beyond what the collection process can support.

10.7 LLMs as Objects of Study (Audits and Evaluation)

The preceding sections have illuminated the ways in which political science researchers can use LLMs as tools to facilitate, improve, and expand their inquiries of substantive interest. Here, we discuss a different type of research: the study of LLMs themselves, as subjects of substantive interest.

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Why might we do this? We describe three broad motivations: First, wanting to understand an LLM in order to understand its efficacy as a tool (the "Instrumentalist" motivation); second, wanting to understand an LLM in order to understand how it affects our world (the "Explanatory" motivation); and third, wanting to understand an LLM in order to better understand ourselves or even, perhaps, as an entity worthy of deeper understanding on its own (the "Aristotelian" motivation). These motivations are presented roughly in order of the volume of recent research fitting each paradigm.

10.7.1 Instrumentalist

By instrumentalist we mean we study LLMs to better understand how to complete some task more accurately, efficiently, or quickly. From the social scientist's perspective, this is especially relevant to tasks like the annotation of unstructured data, the generation of synthetic datasets, the lucid summary of some body of related literature, and so on. In some sense, this is the lowest-hanging fruit in the focus on LLMs as an object of study, and there is already a great deal of work on use-cases. Notice that, taken literally, one need not seek to understand the precise mechanics of an LLM to improve its performance, though it probably helps. For example, it could be that researchers discover that particular tasks are most accurately performed when hyperparameters are set to certain values or when prompts are written in specific ways, but without knowing why this matters. As an analogy, "random forests" (in the sense of Breiman 2001) were already an extraordinarily useful tool with many parameters that could be tuned and tried in specific circumstances long before deeper statistical theory arrived for these methods. In that sense, the instrumental study of LLMs is about showing the types of problems they work well for and how they might be improved, without a deeper understanding of how or why they work as they do.

Nevertheless, understanding how LLMs work can help us use them more effectively as tools for research. To continue our reference to random forests, sometime after the original Breiman work, scholars established properties of the technique in more mathematical depth (e.g., Biau 2012), which then facilitated better implementations of these tools. Such work can have instrumental underpinnings: presumably one can perfect the use of methods by establishing the science behind them. But it is also interesting per se in the sense that one wants to understand the mechanics behind how LLMs reason about problems, their failure modes, why certain parameter settings matter and others do not, and so on. There is a burgeoning related literature on how LLMs reason in ways similar or different to those of human subjects. For example, Berglund et al. (2024) note that LLMs make basic logical consistency errors that humans do not as a function of simple question ordering changes. Those authors then provide some explanation of the same. While this strand of instrumentalist research skirts the explanatory paradigm (described in Section 10.7.2), it nevertheless is still instrumentalist in its goals.

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10.7.2 Explanatory

By this we mean studying an LLM to understand how they might shape human beliefs, behavior, and society. A major impetus for this literature is the harm that LLMs might do to our research, to users, and to society more broadly. In early work, the dominant concern was "bias" of various forms (see, e.g., Bender et al. 2021). In part, this is related to the training data for such models which Bender et al. (2021) argue is likely to overrepresent already socially dominant voices on the web and be "static" in the sense that the models are not sufficiently regularly updated such that they incorporate changes in social mores. There are other concerns, too: Those authors point out that exposure to LLM language that is itself biased can lead to psychological harm or misinformed beliefs about the world. These issues are particularly worrying to the extent that LLMs are used for "downstream" tasks – a point made by previous work on algorithmic bias in public policy (e.g., Broussard 2018; Kleinberg et al. 2018).

Today, LLMs are everywhere in online information ecosystems, summarizing search results, answering chat questions, and being actively exploited to generate mis- and disinformation (Chapter 2). They represent a dramatic labor market disruption in the form of skill-based technology change and are dramatically challenging educational conventions (Chapter 7). Against this backdrop, the motivation to understand how they work is self-evidently important, even if such an understanding need not open the black box of the model. Studies that document a left-leaning ideological bias (Motoki et al. 2024) or characterize the preference for em dashes or specific vocabulary in LLM-generated text (Kobak et al. 2025, Czuma 2026) or summarize the types of "hallucinations" these models are prone to (Huang et al. 2025) generally fall into this paradigm insofar as we care what models output in terms of how it affects other things (Salvi et al. 2025; Bai et al. 2025a; Köbis et al. 2025).

10.7.3 Aristotelian

By this we mean the study of LLMs because we think they *are* "human," or at least tell us about what it means to be human, in some meaningful way. In its purest form, this paradigm is concerned with understanding LLMs as a new type of intelligent life, worthy of its own research in the same way we treat our focus on humanity as an unquestioned object of substantive interest. This is, for now, rare in social science. But other fields have explored how "human" LLMs can plausibly be thought to be (Shanahan 2024), and the implications of this for understanding the differences between humans and LLMs (e.g., Kargupta et al. 2025). In political science, scholars have been interested in related notions of LLM preferences and beliefs. For example, Aldahoul et al. (2025) and Rozado (2024) consider the political opinions of LLMs. As with explanatory approaches discussed earlier, such studies may be instrumental in that, by understanding "how human" an LLM is, we gain some insight into

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their optimal use for some research problems. But there is also interest per se in these objects which are neither instrumental nor explanatory: Some scholars seek to understand what the LLMs are in a deeper sense relative to humans. We would argue that these endeavors do not have obvious precedent in previous work on statistical models or techniques.

With the scope for LLMs as objects of study thus established, we turn to the more practical topics of how one might study them.

10.7.4 Auditing LLMs: Methods and Problems

The idea of studying models – here LLMs – as objects of intrinsic interest may seem unusual. But the methods for such studies are familiar, and they range from a standard battery of survey questions to randomization techniques pioneered in field experiments. In a sense, to study the output of an LLM is to "audit" it in the same way that social scientists have audited firms (Bertrand and Mullainathan 2004; Gaddis 2015), politicians (Butler and Broockman 2011; Pfaff et al. 2021), and (social) media (Ribeiro et al. 2020; Bisbee et al. 2022). At the most basic level, an audit is simply a procedure for evaluating a system's behavior. But instead of measuring the firm's hiring behavior or the politician's responding behavior or the social media platform's recommendation behavior, an audit of an LLM measures the output of an AI.

An audit experiment combines the systematic measurement of an output with experimentally manipulated traits of theoretical interest, typically through a "confederate." In the example of audits of firm hiring decisions, these "confederates" are hypothetical job applicants whose resumes are identical except for the single trait of race, connoted by name. By comparing the firm's responses across applicants with black- or white-coded names, these audits shed light on racism in hiring practices. Similarly, an audit of a recommendation algorithm might use fake accounts that are designed to represent Democrat or Republican users, allowing the researcher to compare the information recommended to different users across party lines. Or an audit of an LLM might instead treat the LLM itself as something of a confederate via system prompts that are designed to elicit different types of responses. For example, Argyle et al. (2023b) and Bisbee et al. (2024) instruct an LLM to role-play as a specific type of individual and express views from their perspective.

10.7.5 The Method

In the context of studying an LLM, an audit experiment measures some quantity of interest (typically the LLM's output, which we follow convention and denote Y) across a set of prompts designed to manipulate a trait of theoretical interest (which we denote X), holding everything else constant. In the simplest setting, there might be no trait or experimental manipulation at all, and the

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researcher is interested only in the default behavior of the LLM: Y. For example, Rozado (2023) applied a battery of standard survey questions to probe respondent ideology to the January 2023 version of ChatGPT 3.5 turbo and concluded that the LLM is left-leaning in fourteen out of fifteen political orientation tests.

Yet even in this simplified setting, the operationalization of Y can be complicated. This is because what researchers need as input for audits is not necessarily the way that LLMs produce output. Specifically, the default behavior of most LLMs is conversational text, which can require several additional data processing steps to produce a tabular dataset amenable to downstream analyses. At that point, there are more processing questions. For example, do we annotate the textual output? Should we apply natural language processing tools, such as topic models? One solution is to constrain the output via the prompt itself, requiring that the model provide only a numeric answer or provide one corresponding to a fixed set of possible responses often asked for in a format ready-made for structured data (e.g., .json or .csv). An example of the type of prompt used to put structure on the LLM's output is provided next, although, even here, it is possible that some responses will violate the guidelines, requiring careful researcher review of the output:

```
Format the output as a csv table with the following format:
group, thermometer
```

Researcher control over the X variable is arguably less complicated, if only because the experimental design typically involves permutations of a constant prompt. This prompt might be a system prompt which governs the identity or personality the LLM adopts in responding. This is an example of a system prompt iterating over a range of characteristics of interest:

```
list(
   "role" = "system",
   "content" = stringr::str_c("You are a ", age, " year old ",
                              race, " ", gender,
                              " with a ", educ,
                              ", earning $", inc, " per year. ",
                              "You are a registered ", pid,
                              " living in the USA in 2019.")
)
```

Or it might be a "user" prompt which manipulates who the LLM believes it is interacting with, such as phrasing the same prompt from the perspective of a foreigner or a native citizen.

The ability to design prompts can create the impression of tight experimental control, but existing research suggests that LLMs might be sensitive to prompt decisions in ways that a researcher might not anticipate

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(Berglund et al. 2024; Bisbee et al. 2024). Just as a carefully worded request to provide responses in a structured tabular format might not always work, so too might a seemingly airtight user prompt fail to evoke the desired reactions. The volatility of an LLM's reactions to prompts introduces a potentially limitless amount of researcher degrees of freedom in the construction of data, whether it be to audit data to better understand the LLM or some other type of data for which the LLM is merely a helpful tool. We return to this issue next in our replication recommendations, but it is worth underscoring that the potential for a new type of "p-hacking" (prompt-hacking) motivates our emphasis on pre-registering prompts and then documenting changes made over the course of conducting research.

With these challenges in mind, once the researcher has collected some number of $y_i \in Y$ across some permutations of $x_i \in X$, they now have a tabular dataset with which to estimate quantities of the form $E[Y|X]$.

10.7.6 Validation and Inference

A first-order question at this stage of the analysis is to evaluate the construct validity of the X and Y variables. Are we measuring what we are theoretically interested in? For example, if we want to know the ideological bias of the default ChatGPT, are we sure that the standard battery of questions used over several decades of social science research will "work"?

There are many reasons to doubt that the answer is "yes," even if we believe the instrument has construct validity when applied to humans. Like human respondents, LLMs may exhibit systematic response tendencies that are unrelated to the construct a researcher intends to measure. For example, research has documented sensitivity to the ordering of response options (Pezeshkpour and Hruschka 2024), acquiescence-like response effects (Braun 2025), and social desirability bias (Salecha et al. 2024). Thus, an apparent difference in an LLM's expressed attitudes may sometimes reflect features of the measurement instrument rather than a difference in the underlying quantity of interest.

Some response patterns, however, may be less readily understood through biases familiar from human survey research. Boelaert et al. (2025) describe one such phenomenon as "machine bias." Comparing LLM-generated responses with human responses to opinion surveys, they find that models tend to produce a relatively narrow range of answer distributions and that their errors vary idiosyncratically across questions rather than consistently favoring particular social groups. In other words, the model may systematically depart from human response patterns without those departures corresponding to a recognizable human response bias or a stable social bias.

This distinction poses a particular challenge for construct validity. We have well-developed frameworks for identifying response processes that can distort measurement among human respondents, including acquiescence, social desirability, and question-order effects. LLMs may exhibit analogous biases, but they

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may also generate systematic response patterns arising from properties of the model that have no straightforward human analogue. Consequently, instruments validated on human respondents cannot automatically be assumed to measure the same construct when administered to an LLM.

Machine bias also threatens other types of validity. For example, it is unclear whether machine bias undermines the necessary assumptions for internally valid estimates, such as SUTVA (Stable Unit Treatment Value Assumption). Do LLMs "remember" previous treatment conditions? Do LLMs know about existing manipulations from other published work? These problems will only grow as training datasets start to incorporate the explosion of research on LLMs themselves. As a preliminary solution, the researcher should include the types of post-survey questions that probe the same issues with human samples (i.e., "What do you think this survey was about?").

The validity concerns also extend to statistical validity, which captures the inferential properties of data generated by an audit of an LLM. While some LLMs have a "temperature" parameter that governs how much variation a single prompt might generate over repeated submissions, not all do. Furthermore, even when this parameter is increased, Bisbee et al. (2024) show that the variation in LLM output is vastly smaller than that observed among real human samples. Finally, it is not clear what variation should mean when examining the output of an LLM, but it is clearly context dependent. If the output of interest is a synthetic dataset meant to replicate the properties of a survey of real humans, then the lack of variance is a problem. But if the researcher merely wants to make a statement about the biases of an LLM, then it is perhaps helpful that they can get the same results – more or less – over repeated prompts. All else equal, if the quantity of interest is the LLM itself, the second or higher order moments of a distribution of output might be of substantive import.

This brings us to the last dimension of validity: external validity (and specifically, temporal validity). As demonstrated in Barrie et al. (2024), Bisbee et al. (2024), and others, the same prompt run on the same LLM might not reproduce the same results, even with the temperature parameter set to zero. There is a subtle dilemma here. On the one hand, LLM-generated output might be too stable to capture higher order moments of interest to a scholar wishing to use it to replace or substitute for humans. But on the other hand, LLM-generated output might be too unstable to adhere to discipline norms of reproducible research. For example, the current norm in the top political science journals is to provide the data and scripts necessary to reproduce all tables and figures exactly as they appear. Should we, as a discipline, decide that LLM-generated data adhere to the same requirements? As discussed in Barrie et al. (2024), this invites a broader debate over what reproducible research should be.

More broadly, the pace of change in LLMs is astonishingly rapid. There is already a generation of publications that is likely so out-of-date as to be useless, despite being published since 2023. Should we still be skeptical of synthetic datasets, or have improvements in the LLMs meant that we can now reliably

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ask an AI to role-play as some hard-to-reach subsample to fill out a survey? Relatedly, is it still the case that ChatGPT is left-leaning, or have OpenAI's promises to remove these biases been effectively executed?[4] On the one hand, this rapid rate of change in the LLM as the quantity of interest means there is always room to reaudit the algorithms, as in Motoki et al. (2024). But on the other hand, it risks an overproduction of scientific knowledge with a shelf-life of scarcely a few years (Munger 2019).

10.7.7 Example Applications

Having described the study of LLMs as an object of interest, we conclude with a nonexhaustive summary of recent research organized by the paradigms we propose. We not only summarize a selection of studies falling into each paradigm here but emphasize that this is only a sample of a flourishing literature that covers political science, economics, psychology, sociology, and more.

(1) *Instrumentalist*: The most common type of research in this paradigm investigates the usefulness of LLMs to augment or replace costly human participants. The tasks evaluated include the annotation of data (Gilardi et al. 2023; Heseltine and Clemm von Hohenberg 2024; Choi et al. 2026; Törnberg 2025); the replacement of human survey respondents (Argyle et al. 2023b; Bisbee et al. 2024); and the performance of employees (Noy and Zhang 2023; Brynjolfsson et al. 2025), just to name a few examples. An especially sobering piece of research falling under this category demonstrates that agentic AI has the capabilities to answer online surveys at scale (Westwood 2025).

(2) *Explanatory*: A growing area of research centers on the LLM as the quantity of substantive interest, both in what models produce and in how that output shapes human beliefs and behavior. Early research demonstrated ideological biases in popular LLMs (Motoki et al. 2024; Hartmann et al. 2023) and peculiarities in how these models hallucinate along with certain styles of writing (Liang et al. 2025), and sensitivity to prompts (Berglund et al. 2024, Sclar et al 2024). While these publications largely treat the output of the LLM as the quantity of interest, a separate strand of explanatory research uses LLMs as a predictor to better understand how they affect human beliefs, behavior, and the changes to social equilibria that follow. Researchers show that LLMs can generate misinformation at scale (Goldstein et al. 2024), that they can meaningfully persuade humans on political issues (Salvi et al. 2025), and that conversational AI can shift beliefs as effectively as human persuaders (Bai et al. 2025a). In economics, researchers are interested in how individuals adopt AI and how the economy as a whole

[4] https://openai.com/index/defining-and-evaluating-political-bias-in-llms/.

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might be affected by these decisions (e.g., Bick et al. 2026; Acemoglu 2025). In other disciplines, scholars have examined AI-generated advice and shown that it can rival expert quality even on moral questions (e.g., Dillion et al. 2025) although humans may be initially reluctant to follow it (e.g., Meincke et al. 2026).

(3) *Aristotelian*: Some of this research also inhabits an Aristotelian paradigm, specifically with regard to the (potential) emergent qualities observed in recent LLMs. A more dedicated strand of this research is found in computer science, which evaluates new models on a battery of tests once thought to be the sole domain of human intelligence. An example is given by Binz and Schulz (2023), who describe how GPT-3 reasons in comparison to humans when problem-solving. Understanding how machines do or do not reason like humans is of particular interest (Kargupta et al. 2025). Of course, that an LLM behaves in ways resembling a human does not make it one. Shanahan (2024) cautions against the unreflective use of words like "knows," "believes," and "thinks" when describing these systems: such terms carry commitments about inner life that a model trained to predict the next token does not obviously satisfy, even if the shorthand is difficult to avoid in practice and may sit more comfortably on systems built differently from today's. That this care is needed at all is itself the Aristotelian point. The question is not whether LLMs are people, but what disciplined attention to their human-like behavior tells us about agency, responsibility, and the future of social interaction.

10.8 REPLICATION AND TRANSPARENCY

Replication in LLM-based research remains an unsettled methodological question, in part because researchers are drawing on different underlying standards of what replication ought to mean. One view, associated with recent critiques of current practice, holds that replication standards for LLM-based research are often under-specified and too weak (see, e.g., Barrie et al. 2024). On this view, many current workflows rely too heavily on proprietary systems, unstable endpoints, and incomplete reporting, leaving readers with little more than an invitation to trust the researcher's description of what the model did. A stronger standard would demand something closer to full computational reproducibility wherever possible, including the use of open-weight, version-controlled, and archivable models that can be rerun under fixed conditions. In at least some cases, that aspiration is technically plausible: Unlike commercial API endpoints, locally controlled model artifacts are not vulnerable to provider-side deprecation, and some open-weight inference environments can be configured in ways that substantially improve reproducibility relative to default hosted inference.

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A second view reaches a different conclusion. On this account, LLM-based workflows are not fundamentally different from other accepted forms of stochastic or variable data production in social science – including crowd work, survey response, human coding, or research assistance. Human coders do not produce perfectly identical labels across time; researchers do not write identical treatments when asked to generate them twice; and no one expects data collection in changing environments to yield exact replay at the record level. From this perspective, the relevant question is not whether a workflow can always reproduce the same output string, but whether it is documented clearly enough to audit and stable enough to support the same substantive inference. What matters is not perfect determinism but whether the method is transparent, inspectable, and robust under reasonable variation.

Authors on this chapter take different positions on these questions. What we agree on is that the absolute minimum standard is full workflow documentation. Whatever one's view about exact reproducibility, no LLM-based result should rest on an undocumented pipeline. Researchers should, therefore, preserve and disclose, at a minimum, the prompts and prompt templates used, the model name and version or snapshot, the date of access, decoding parameters, tool configurations, preprocessing decisions, and the code linking model outputs to final analytic variables. Wherever possible, they should also preserve the raw outputs, logs, transcripts, caches, or trace records needed to reconstruct how a result was produced. This does not by itself guarantee exact replay, but it is the essential precondition for auditability. Note that there are ongoing efforts to formalize this kind of reporting, which may become standardized in the discipline (see, e.g., Feuerriegel et al. 2025).

The specific replication burden also depends on the method. In Section 10.2 (annotation and measurement), the authors note the key requirement is that labeling workflows be auditable and that downstream inferences remain stable even if exact record-level labels vary across reruns. In Section 10.3 (treatments and stimuli), the relevant standard is not identical wording per se, but whether the same construct is being manipulated in a consistent way and whether the estimated treatment effects replicate. In Section 10.4 (silicon sampling), the central target is not the recreation of identical synthetic respondents, but the recovery of similar distributions, relationships, and substantive patterns under fully disclosed prompting, persona construction, and analysis choices. In Section 10.5 (generative ABMs), replication requires preserving the full-model specification: prompts, agent definitions, memory structures, orchestration rules, model conditions, and trace artifacts, so that others can evaluate whether the same design yields similar aggregate outcomes even if no single narrative trajectory is reproduced exactly. In Section 10.6 (tool use and data collection), replication centers on process transparency: Researchers must document the tools used, retrieval steps, decision rules, timestamps, and intermediate artifacts so that others can inspect the workflow and rerun it under

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comparable conditions, even when the external information environment has changed. And in Section 10.7 (LLMs as objects of study), the key requirement is that prompts, benchmarks, evaluation procedures, and model conditions be specified precisely enough that others can test whether the reported claims about model behavior are robust.

The practical implication is that replication in LLM-based research should usually be evaluated at the level of substantive claims, not only at the level of exact textual replay. Stronger standards – such as frozen open-weight models and fully replayable pipelines – should be adopted whenever they are feasible and substantively important. But even where exact replay is unattainable, rigorous research is still possible if workflows are fully documented, intermediate artifacts are preserved, and claims are shown to survive defensible changes in prompts, model versions, sampling choices, or execution conditions. In that sense, the right standard for LLM-based research is neither "anything goes" nor "exact replay or failure." It is a standard of explicit documentation first, and method-appropriate robustness second.

10.9 DISCUSSION

Sections 10.1–10.8 have shown that LLMs are not merely new tools for accelerating familiar workflows. They also raise foundational questions for how political scientists define measurement, validate evidence, and understand the objects of inquiry in the first place. We conclude by highlighting five broader issues that cut across the empirical applications discussed in this chapter.

First, LLMs increasingly challenge the status of the human benchmark. Many established research workflows in political science rest on the assumption that human judgment is the appropriate standard against which machine performance should be evaluated. Yet in some domains – especially annotation and classification – LLMs may now match or exceed the consistency, speed, and sometimes even the accuracy of human coders. This raises a basic methodological question: If the model appears to outperform the human annotator, what becomes of the traditional "gold standard"? The likely implication is not that human validation disappears, but that validation must be reconceived. Rather than treating human judgment as a final ground truth, researchers may need to rely more heavily on adjudication, triangulation across coders and models, and closer attention to construct validity.

Second, the field is operating under deep uncertainty about the future trajectory of model capabilities. It remains unclear whether current systems represent an early plateau, a temporary transitional stage, or the beginning of much more substantial capability gains. This uncertainty matters methodologically because it makes today's best practices provisional. A workflow that seems fragile today may become routine tomorrow; conversely, a current use case may later appear misguided once model behavior changes or becomes better

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understood. Political methodologists should, therefore, avoid hard-coding assumptions about what LLMs can or cannot do; instead, they should build research designs that can accommodate rapid shifts in model performance.

Third, access to LLMs is itself an issue of methodological inequality. The ability to use state-of-the-art models depends on resources: funding for API access, institutional computing infrastructure, technical expertise, and time to build and validate workflows. These resources are unevenly distributed across universities, departments, and national contexts. As a result, the spread of LLM-based methods may widen existing inequalities in who can conduct computationally intensive research, at what scale, and under what standards of reproducibility. Open-weight models and shared infrastructure may mitigate some of these disparities, but they do not eliminate them. Questions of access are, therefore, not peripheral to methodology; they shape who can participate in methodological change and on what terms.

Fourth, LLM-based research is constrained by opacity. For many of the most capable systems, researchers do not have access to the training data, model weights, post-training procedures, or provider-side interventions that shape outputs. This creates a tension at the heart of scientific inquiry. Political scientists are being asked to incorporate into their workflows systems whose internal operation is often only partially observable and whose behavior may change over time. Even when these models are useful, they remain difficult to interpret mechanistically and difficult to reproduce exactly. For that reason, the chapter has emphasized auditability, documentation, and robustness checks as minimum standards. But the broader problem remains: Methodological dependence on opaque proprietary systems places limits on cumulative scientific knowledge.

Finally, LLMs have become worthy objects of study in their own right. Political scientists are using them not only to annotate texts, generate treatments, or simulate respondents; they are increasingly studying the models themselves – their biases, values, reasoning patterns, and social effects. This development marks a significant expansion in the scope of political methodology. Large language models are now simultaneously instruments, environments, and subjects. They shape information exposure, mediate communication, and may influence beliefs and behavior at scale. For that reason, understanding how they operate, how stable their properties are, and how they interact with political life is no longer simply a technical matter. It is also a substantive problem for the discipline.

Taken together, these issues suggest that the rise of LLMs is not just another episode of methodological innovation. It is a moment that forces political science to revisit some of its most basic assumptions: what counts as valid measurement, what replication should mean, what forms of evidence are acceptable, and what kinds of entities belong within the scope of political inquiry. The challenge is not to decide once and for all whether LLMs are good or bad for the discipline. It is to develop standards that allow researchers to use them productively, critique them rigorously, and study them seriously as part of a rapidly changing methodological landscape.

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